



MCS3301 – Control Engineering I

Dr. Ado Haruna

Dept. of Mechatronics Engineering

Bayero University, Kano

aharuna.mct@edu.buk.ng



Steady State Gain

- Open loop system responses are a precursor to analyzing closed loop system behavior.
- A general observation can be made about the **steady state** behavior of systems.
- Consider the system $Y(s) = G(s)U(s)$ such that $u_{ss} = \lim_{t \rightarrow \infty} u(t)$ exists.
- The final value theorem states that $u_{ss} = \lim_{s \rightarrow 0} sU(s)$
- If this limit exists then
$$\begin{aligned} y_{ss} &= \lim_{t \rightarrow \infty} y(t) \\ &= \lim_{s \rightarrow 0} sY(s) \\ &= \lim_{s \rightarrow 0} sG(s)U(s) \\ &= G(0)u_{ss} \end{aligned}$$
- $G(0)$, if it exists, is called the **steady state gain**.



First Order Systems

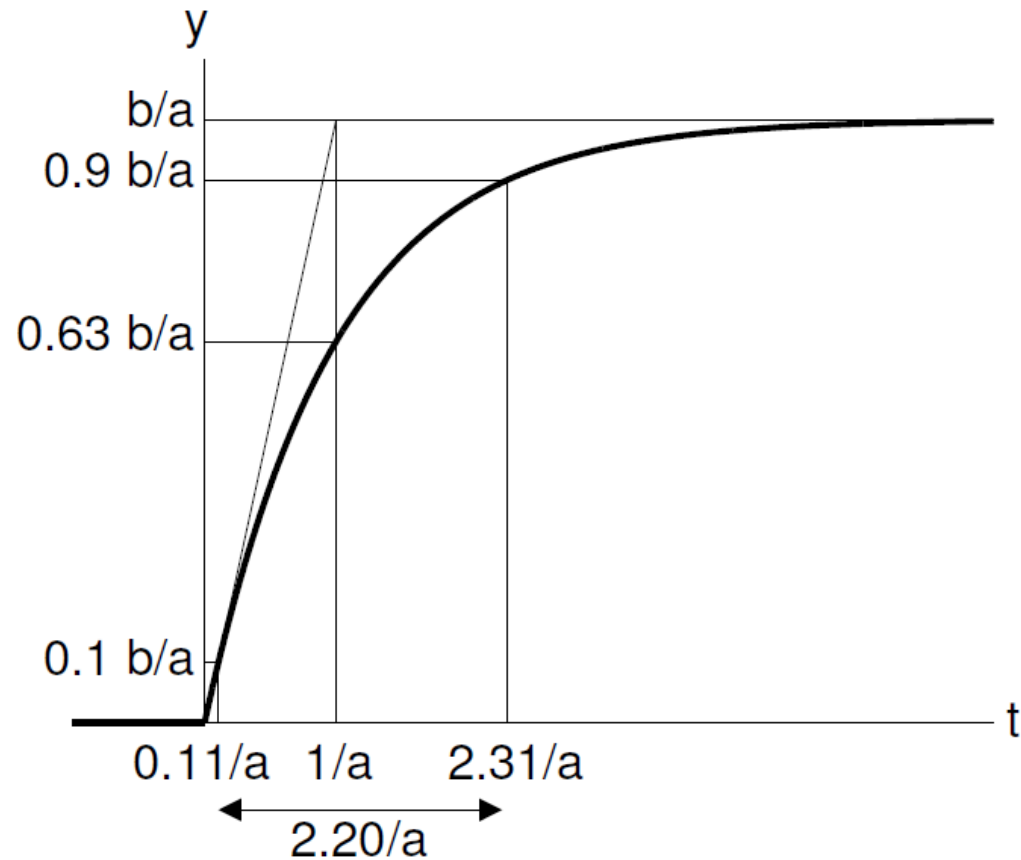
- A first order system has the transfer function $G(s) = \frac{b}{s+a}$
- The steady state gain is b/a .
- Its time response can be derived from Laplace transforms

$$Y(s) = G(s)U(s) = \frac{b}{s+a} \cdot \frac{1}{s} = \frac{b}{a} \left(\frac{1}{s} - \frac{b}{s+a} \right)$$

- For $t \geq 0$, $y(t) = \frac{b}{a} (1 - e^{-at})$
- The response reaches approximately 63.21% of its final value at $t = \frac{1}{a}$ referred to as the **time constant**.
- The response reaches 10% of its final value after $t = 0.1054/a$ and 90% of its final value after $t = 2.306/a$
- The difference between these two values is referred to as the **rise time** by some authors while others take the time between 0% and 90%.
- If a is negative then $y(t)$ blows up to infinity and the response is said to be **unstable**.



First Order Step Response





Second Order Systems

- A second order system transfer function can be expressed in the form

$$G(s) = \frac{Y(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

- Where ω_n is the **natural frequency** and ζ is the **damping ratio**
- The characteristic equation is obtained as

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

- For a unit step function, $R(s) = 1/s$ and the **output response** can be obtained by taking the inverse Laplace transform which gives

$$Y(s) = 1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1 - \zeta^2}} \sin\left(\omega_n \sqrt{1 - \zeta^2} t + \cos^{-1} \zeta\right)$$



Second Order Systems

- The second order system characteristic equation can also be expressed as

$$as^2 + bs + c = 0$$

- The roots of the characteristic equation (poles of the system) determine the transient behaviour and are given by

$$p_1, p_2 = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Roots	Damping ratio	Transient response
p_1 and p_2 are –ve real and unequal	$\zeta > 1$	Overdamped
p_1 and p_2 are –ve real and equal	$\zeta = 1$	Critically damped
p_1 and p_2 are complex conjugates of the form $\sigma \pm j\omega$	$0 < \zeta < 1$	Underdamped
p_1 and p_2 are imaginary of the form $\pm j\omega$	$\zeta = 0$	Undamped



Second Order Systems: Real Poles

- The system can be expressed as

$$G(s) = \frac{p_1 p_2}{(s + p_1)(s + p_2)}$$

- If either p_1 or p_2 is **negative** (i.e. the corresponding pole is positive) then the step response blows up i.e. the system is **unstable**.
- If both p_1 and p_2 are **positive**, then the system is **stable**.
- If $p_1 \gg p_2$ then the step response may be approximated as $y(t) \approx 1 - e^{-p_2 t}$ and the response is similar to a **first order** system with a pole at p_2
- If p_1 and p_2 are similar, the system is **critically damped** and the response is significantly different from any 1st order response.



Second Order Systems: Complex Poles

- When $0 < \zeta < 1$ the poles form a **complex conjugate** pair that lie at

$$p = \omega_n \zeta \pm j \omega_n \sqrt{1 - \zeta^2} = \sigma \pm j \omega$$

where $\sigma = \omega_n \zeta$ and $\omega = \omega_n \sqrt{1 - \zeta^2}$

- Recall that the output response is given by

$$Y(s) = 1 - \frac{e^{-\zeta \omega_n t}}{\sqrt{1 - \zeta^2}} \sin \left(\omega_n \sqrt{1 - \zeta^2} t + \cos^{-1} \zeta \right)$$

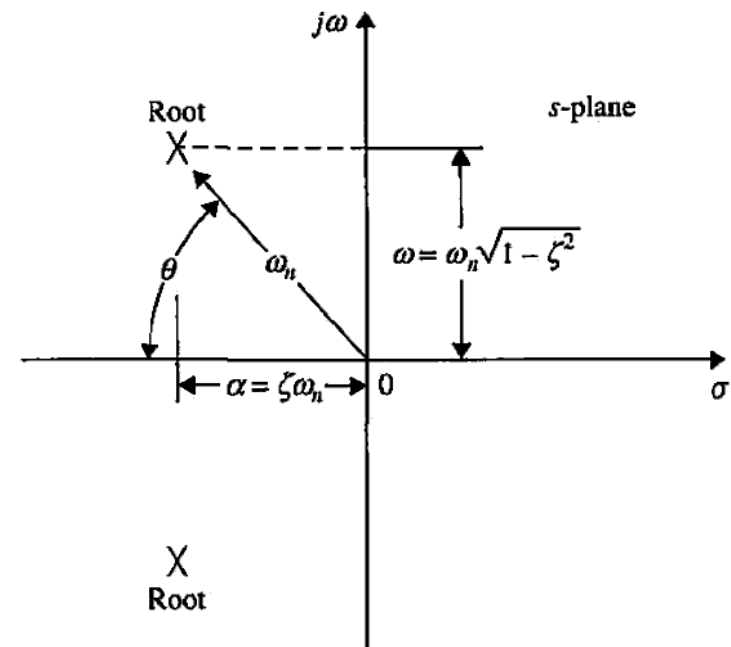
- σ appears as the constant that is multiplied by t in the exponential term of $y(t)$.
- Therefore, σ controls the rate of rise or decay or **damping** of the unit-step response $y(t)$ and is known as the **damping constant**.
- When the system is critically damped i.e. $\zeta = 1$, the damping factor is $\sigma = \omega_n$
- Thus ζ is regarded as the **damping ratio** defined as

$$\zeta = \frac{\text{actual damping factor}}{\text{damping factor at critical damping}}$$



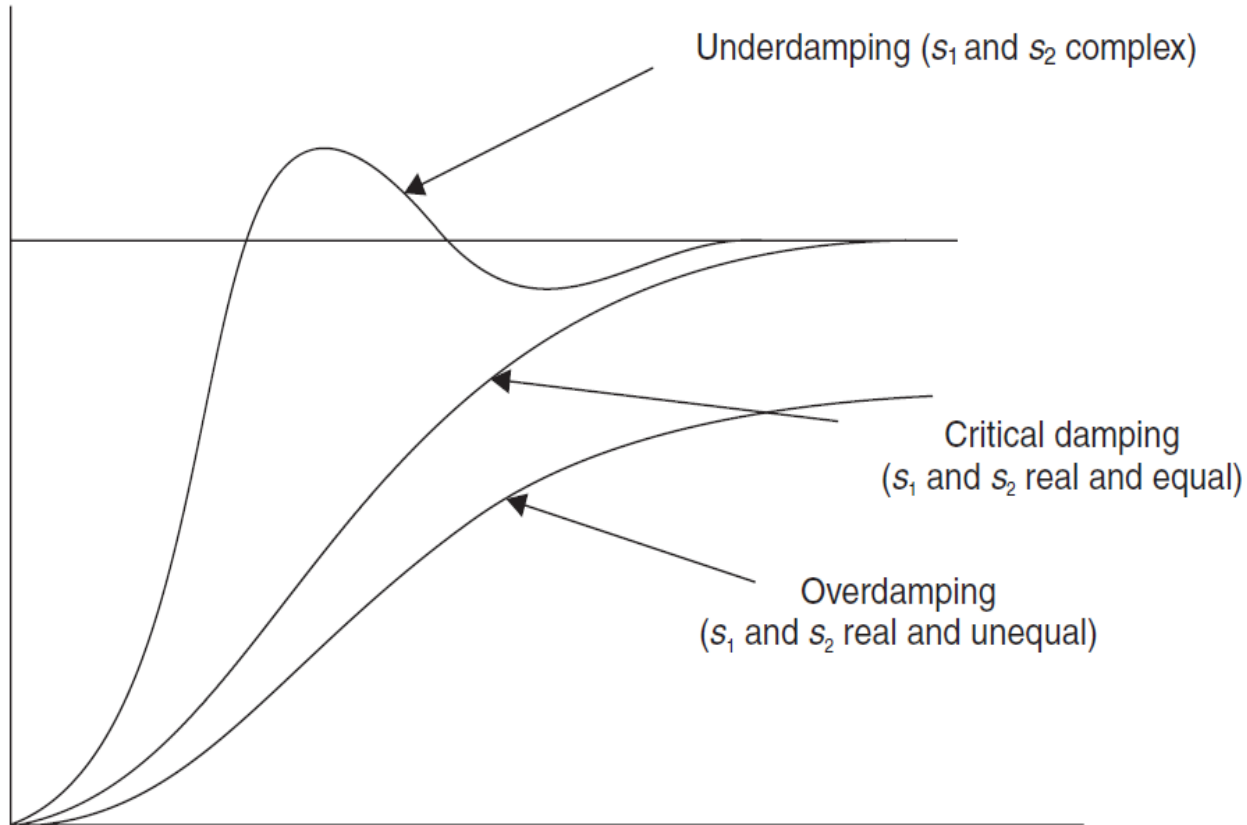
Second Order Systems: Complex Poles

- The ω_n parameter is defined as the **undamped natural frequency**.
- From the expression $p = \omega_n \zeta \pm j\omega_n \sqrt{1 - \zeta^2}$, when $\zeta = 0$ the **poles** of the system are purely **imaginary**.
- The equation of the output response shows the step response is purely sinusoidal when $\zeta = 0$.
- Therefore ω_n corresponds to the frequency of the undamped sinusoidal response.
- It is also seen from $\omega = \omega_n \sqrt{1 - \zeta^2}$ that when $0 < \zeta < 1$, the imaginary part of the roots has a **magnitude** of ω .
- ω_n is the **radial distance** from the pole to the origin of the s plane.
- σ is the **real part** of the roots
- $\zeta = \cos \theta$





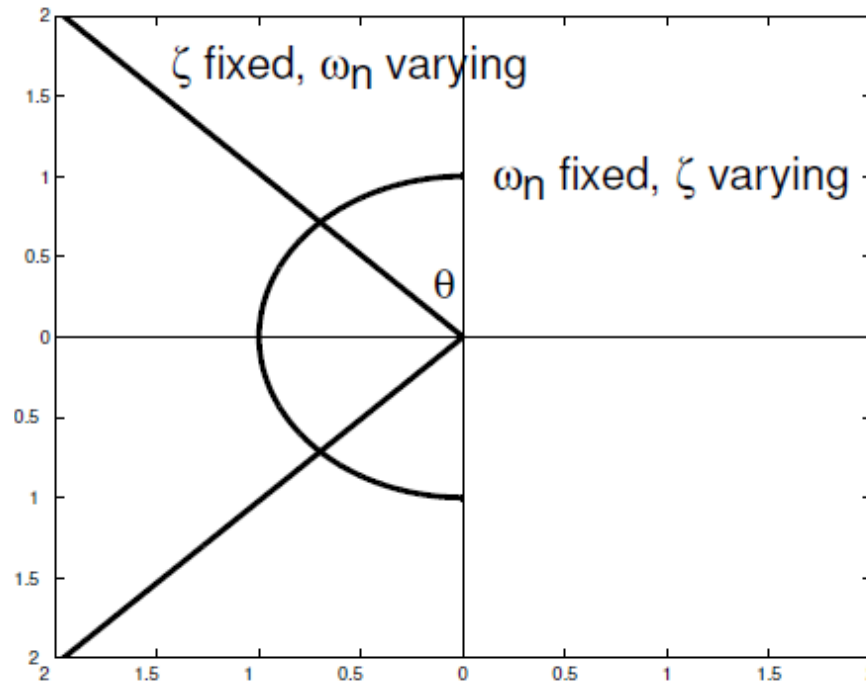
Second Order Systems





Second Order Systems

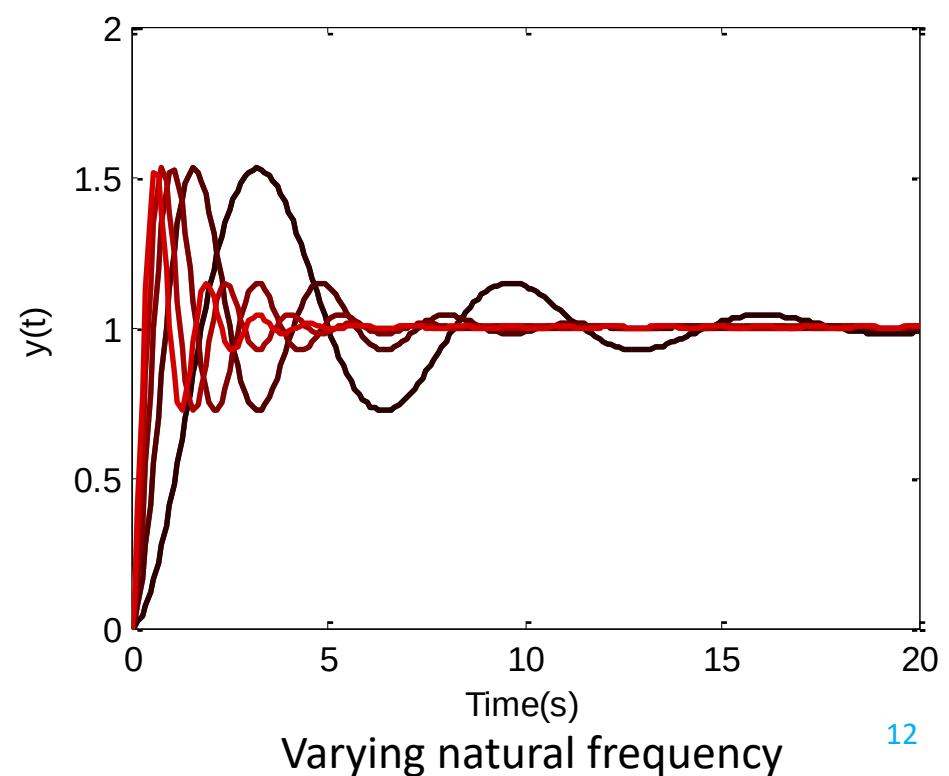
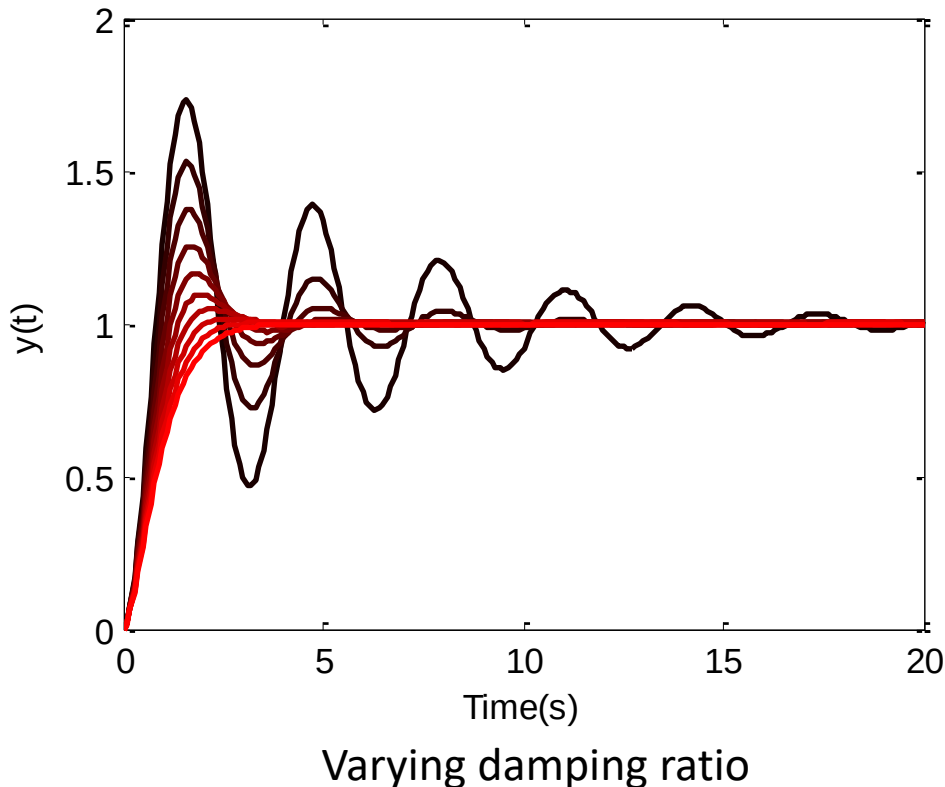
- A locus of the poles of a typical second order system is shown below for a varying ω_n with a fixed ζ and a varying ζ with a fixed ω_n .





Second Order Systems

- As ζ decreases (red to black), the **overshoot** increases.
- When $\zeta > 1$, (pure red line) the step response does **not** exhibit any **overshoot** i.e, $y(t)$ never exceeds its final value during the **transient**.
- The responses also show that ω_n has a **direct** effect on the **rise time** and **settling time** but does not affect the overshoot.





Time Domain Performance Characteristics

- The **Rise Time** is the time taken for the output to go from about 10% to 90% of its final value just as for first order systems
- The **Settling Time** is the time for which the response reaches to within 1% of the final value and then remains within that boundary.
- The **Peak Overshoot**, M_p , for a stable response with a well-defined final value $y_{ss} = \lim_{t \rightarrow \infty} y(t)$ is given by

$$M_p = \frac{y_{max} - y_{ss}}{y_{ss}}$$

- The overshoot is often expressed as a percentage i.e.

$$OS\% = \frac{y_{max} - y_{ss}}{y_{ss}} \times 100$$

- For underdamped second order systems, the peak overshoot occurs at

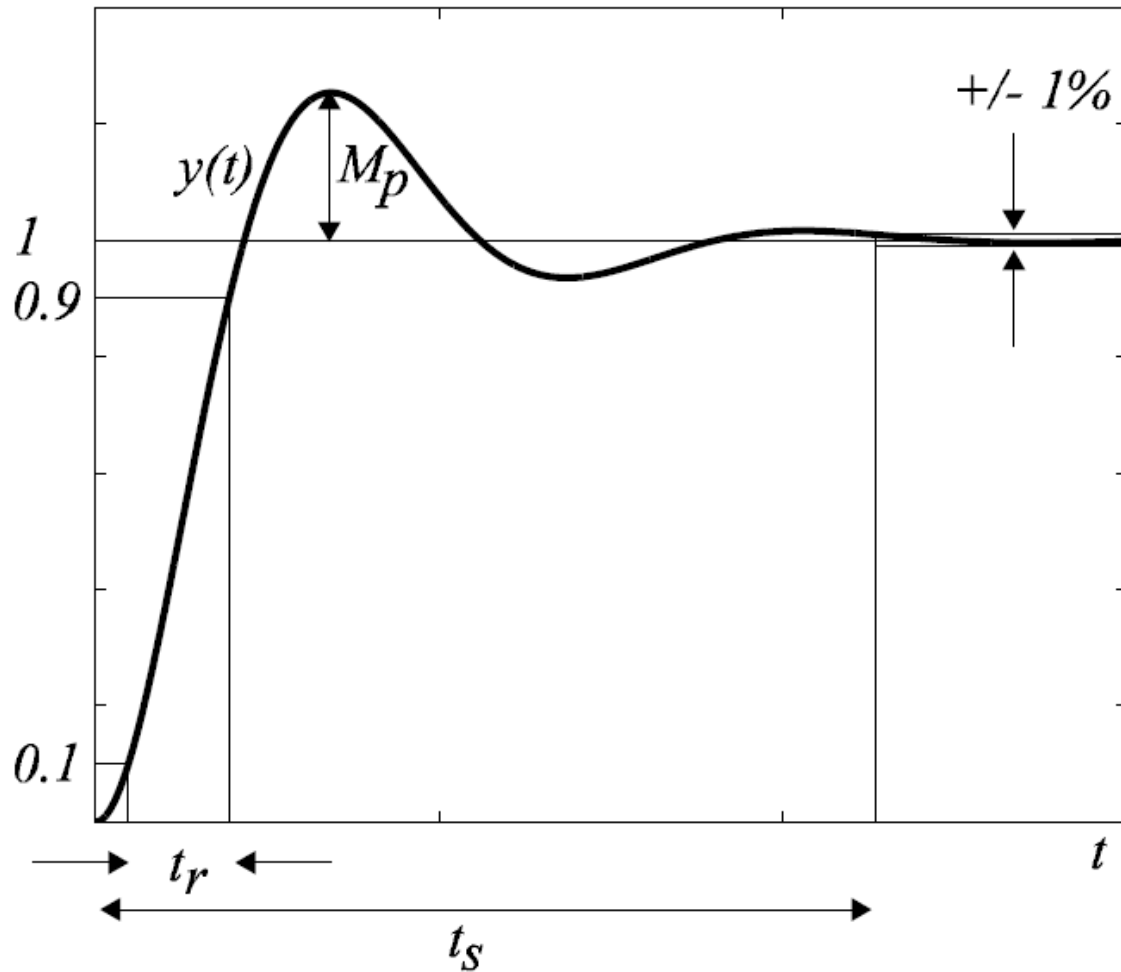
$$t = \frac{\pi}{\omega_n \sqrt{1 - \zeta^2}}$$

and

$$M_p = e^{-\frac{\zeta}{\sqrt{1-\zeta^2}}\pi}$$



Time Domain Performance Characteristics



$t_r = \text{rise time}$

$t_s = \text{settling time}$

$M_p = \text{peak overshoot}$